

Mathematics for Machine Learning

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Definition

Let $f : X \rightarrow \mathbb{R}$, $X \subset \mathbb{R}$. It is said f is differentiable at interior point $x_0 \in X$, if the following limit exists

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Proposition

$$f(x_0 + h) - f(x_0) = f'(x_0)h + o(h), h \rightarrow 0.$$

Derivative

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Theorem

If f has a finite derivative at x_0 then it is continuous at x_0 .

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$$\textcircled{4} \quad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}, \text{ if } g \neq 0,$$

$$\textcircled{5} \quad (fg)^{(n)} = \sum_{k=0}^n C_n^k f^{(k)} g^{(n-k)}.$$

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Theorem

Let $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$.

Theorem

Let $f, g : [a, b] \rightarrow \mathbb{R}$ are continuous on $[a, b]$, differentiable on (a, b) and $g'(x_0) \neq 0, x \in (a, b)$. Then there exists $c \in (a, b)$ such that

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

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Theorem

Let $f : X \rightarrow \mathbb{R}$ and X is interval. f is increasing (decreasing) on X if and only if $f'(x) \geq 0$ ($f'(x) \leq 0$) for all $x \in X$.

L'Hospital's rule

Theorem

If

- the functions f and g are differentiable in (a, b) and $g(x) \neq 0, x \in (a, b)$,
- $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$,
- $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = K$,

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Theorem

If

- the functions f and g are differentiable in (a, b) and $g(x) \neq 0, x \in (a, b)$,
- $\lim_{x \rightarrow a} g(x) = +\infty$,
- $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = K$,

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Definition

Let $f : X \rightarrow \mathbb{R}$ and x_0 is an interior point of X . Then x_0 is called local maximum (minimum) point of f , if there exists $\delta > 0$ such that from $x \in (x_0 - \delta, x_0 + \delta)$ follows that $f(x) \leq f(x_0)$ ($f(x) \geq f(x_0)$).

Extrema of Function

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Theorem

If x_0 is an extremum point of f and there exists $f'(x_0)$, then $f'(x_0) = 0$.

Theorem

Let f is differentiable in the intervals $(x_0 - \delta, x_0)$, $(x_0, x_0 + \delta)$ and continuous at x_0 , then

- 1 if $f'(x) > 0$, $x \in (x_0 - \delta, x_0)$ and $f'(x) < 0$, $x \in (x_0, x_0 + \delta)$, then x_0 is a local maximum point,*

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- ❷ *if $f'(x) < 0$, $x \in (x_0 - \delta, x_0)$ and $f'(x) > 0$, $x \in (x_0, x_0 + \delta)$, then x_0 is a local minimum point,*
- ❸ *if $f'(x)$ doesn't change it's sign then x_0 is not an extremum point.*

Theorem

Let $f'(x_0) = 0$ and there exists finite $f''(x_0)$, then

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Definition

The point x_0 is called a saddle point of function f , if there exists $\delta > 0$ such that the tangent line of the graph of the function f at the point $(x_0, f(x_0))$ lies in different sides of the graph in the intervals $(x_0 - \delta, x_0)$ and $(x_0, x_0 + \delta)$.

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Theorem

Let f be a twice differentiable function at x_0 . If there exists $\delta > 0$ such that f'' has different signs in the intervals $(x_0 - \delta, x_0)$ and $(x_0, x_0 + \delta)$, then x_0 is a saddle point of function f .

Definition

Let $f : X \rightarrow \mathbb{R}$ and $X \subset \mathbb{R}$ is an interval. The function f is called convex if

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y)$$

for all $x, y \in X$ and $\alpha \in [0, 1]$.

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Theorem

Let $f : X \rightarrow \mathbb{R}$ and $X \subset \mathbb{R}$ is an interval. If f is a convex function, then

$$f(\alpha_1 x_1 + \dots + \alpha_n x_n) \leq \alpha_1 f(x_1) + \dots + \alpha_n f(x_n),$$

for all $x_i \in X$, $\alpha_i \in [0, 1]$, $1 \leq i \leq n$ such that $\sum_{i=1}^n \alpha_i = 1$.

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Definition

The series $\sum_{n=1}^{\infty} a_n$ is called convergent if A is finite, otherwise it is called divergent.